



Aces Simulation

A simulation that allows you to be in control and check how much the pandemic is affecting climate change. <https://digit.in/aug20-53>



Atmospher rings like a bell

Researchers find that there are large scale resonant waves in the air that move across the planet. <https://digit.in/aug20-54>

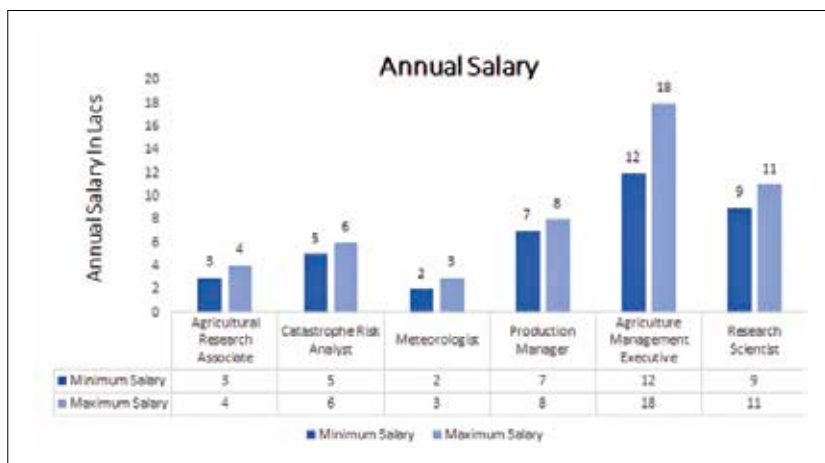
loons, radar systems, satellites and sensors. These tools are used to collect data while computer programs analyse it.

Atmospheric scientists must be well-versed with these instruments, programs, and software and therefore, they need to be somewhat tech-savvy. They must also be able to grasp new methods of data collection and analysing quickly since the technology in the field keeps evolving. These instruments are absolutely critical to understanding air pollution, drought, loss of ozone cover, and other atmospheric issues. Atmospheric scientists also make use of graphics software to illustrate their forecasts and reports. Some atmospheric scientists even work very closely with geoscientists or even social scientists to solve issues in areas such as commerce, energy, agriculture, transportation, and the environment.

THE PATH TO BECOMING AN ATMOSPHERIC SCIENTIST

Aspiring atmospheric scientists must strive to get a bachelor's degree in meteorology or other closely-related earth sciences. If you want to get into the more research-intensive side of atmospheric science, then you will need a master's degree at minimum, but holding a PhD is ideal. Other than subjects relating to meteorology and earth sciences, advanced courses in physics and mathematics is also highly advised. Also, classes in oceanography and geophysics are commonly recommended.

Additionally, special courses in computer programming could also prove to be highly effective since some atmospheric scientists may have to write and edit computer programs to produce forecasts. Some organisations are even asking for candidates to have some background in studying communication since atmospheric scientists need to be able to communicate effectively to make their data accessible to the public. In India, meteorology is



The jobs across different areas of atmospheric science in India and their average annual salary

the most common field of atmospheric science that people get into, and there are plenty of courses available to that end. Candidates can pursue an M.Sc in Meteorology, a Diploma in Meteorology and Atmospheric Sciences, a B.Tech or M.Tech in Meteorology and even a PhD in the field. Some universities that offer meteorology courses include IIT Kharagpur, IIT Delhi, Indian Institute of Tropical Meteorology, Bharathiyar University, Punjabi University, Cochin University of Science and Technology, and MS University of Baroda.

SCOPE FOR GROWTH

Most of the job opportunities for atmospheric scientists in India are in the public sector and not the private sector. Our government, namely the Ministry of Earth Sciences recruits atmospheric scientists for various scientific and contractual posts. Mostly, meteorology, a branch of atmospheric science, has the most demand in India. Individuals who com-

plete their education in atmospheric and earth sciences, and meteorology can join meteorological departments across the country such as Nuclear Meteorology, Maritime Meteorology, Hydrometeorology, Weather Forecasting, Aviation Meteorology, Agricultural Meteorology, Environment Agency,

Service industries, Fisheries Sector, Environmental Consultancy and more. They can even become professors in some colleges and universities that offer such courses.

While there are quite a few jobs for atmospheric scientists in India to get into, the salary is slightly on the lower side since most of them are government jobs. However, if you get into less-common fields and go higher up in ranks, the salary can get considerably high. Nevertheless, if you choose to move to another country to pursue this field, the salary prospects are higher. In the US, the median salary for atmospheric scientists is about \$95,380, with the highest being \$147,160. The demand for atmospheric scientists is expected to grow by 12.5 per cent or more between 2016 – 2026 in the US. If you are interested in the field, here are some new books relating to the subject that could help you get a better understanding of the field and the study –

- Atmospheric Science for Environmental Scientists – C. Nick Hewitt, Andrea V. Jackson
- Probability, Statistics, And Decision Making In The Atmospheric Sciences – Allan Murphy, Richard W. Katz
- The Weather Machine – Andrew Blum
- Statistical Methods in the Atmospheric Sciences – Daniel S. Wilks
- Understanding Climatology – Salvador Poole



The IMD (Indian Meteorological Department) is home to many meteorologists